

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

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| <p>A) What are Maxwell's equations, and how do they relate to the behavior of electric and magnetic fields?</p> | Understand | 6 |
| <p>B) Given two points C (-4,2,1) and D (5, 20⁰, -80⁰) Find
(i) Spherical co-ordinates of C (ii) Cartesian co-ordinates of D (iii) the distance from point C to D</p> | Understand | 6 |
| <p>C) Consider two vectors A and B, given by:
 $\mathbf{A} = 3\mathbf{a}_x + 2\mathbf{a}_y - 5\mathbf{a}_z$
 $\mathbf{B} = \mathbf{a}_x + 4\mathbf{a}_y + 2\mathbf{a}_z$
 Find a) $\mathbf{A} + 3\mathbf{B}$, (b) $\mathbf{A} \cdot \mathbf{B}$ and (c) angle between A and B.</p> | Understand,
Remember | 6 |

Q.2 Solve Any Two of the following.

12

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| <p>A) Show that the net power flowing out of a given volume v is equal to the time rate of decrease in the energy stored within volume v minus the ohmic losses.</p> | Understand,
Remember | 6 |
| <p>B) Explain Poincare's Sphere and its significance in understanding wave polarization</p> | Understand,
Remember | 6 |
| <p>C) The Electric field amplitude of a uniform plane wave propagating in $\overline{az}$ direction is 250 V/m. If $\overline{E} = E_x \overline{ax}$ and $\omega = 1.00$ Mrad/sec Find The frequency, Wavelength and Time period.</p> | Understand | 6 |

Q. 3 Solve Any Two of the following.

12

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| <p>A) A transmission line cable has following primary constants. $R = 80 \Omega/\text{km}$, $G = 55 \mu\text{mho}/\text{km}$, $L = 1.8 \text{ mH}/\text{km}$ and $C = 0.075 \mu\text{F}/\text{km}$, at a signal of 3 kHz, calculate (i) characteristics impedance, (ii) attenuation constant (iii) phase constant (iv) wavelength (v) velocity and (vi) time taken by signal to travel 100 km along line.</p> | Understand | 6 |
| <p>B) How is impedance matching achieved with stubs? Distinguish between single and double stub matching in a transmission line</p> | Understand | 6 |
| <p>C) Define / Explain the following terms
 (a) Propagation constant, (b) Characteristic impedance, (c) Reflection Coefficient, (d) VSWR and (e) Smith Chart</p> | Understand | 6 |

Q.4 Solve Any Two of the following.		12
A) Explain the concept of TE (Transverse Electric) and TM (Transverse Magnetic) modes in a rectangular waveguide. How are the electric and magnetic field components distributed in each mode?	Understand	6
B) A rectangular waveguide with dimensions $a = 2.5$ cm and $b = 1$ cm is operating at 20 GHz. (a) Find the cutoff frequency for the TE ₂₀ mode. (b) The guide wavelength for the TE ₁₀ mode.	Understand, Remember	6
C) How does wave polarization change when a wave crosses a media interface with different refractive indices?	Understand	6
Q. 5 Solve Any Two of the following.		12
A) Describe the radiation parameters of an antenna, such as radiation pattern, directivity, and gain. How are these parameters related to the antenna's performance in transmitting or receiving signals?	Understand, Remember	6
B) Derive an expression to find put fields (E and H) radiated by Hertzian dipole.	Understand	6
C) Compare the radiation characteristics of a monopole antenna and a dipole antenna. How does their radiation pattern differ, and what are their respective applications?	Understand	6

***** End *****